CS 216: Bitcoin Transaction Assignment

Team Information

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GitHub Repository:

https://github.com/sairaj22019/Bitcoin-Transaction-Assignment.git

Project Objective and Aim

The primary aim of this project is to practically explore the mechanics of creating, signing, and validating Bitcoin transactions across two distinct address formats: Legacy (P2PKH) and SegWit (P2SH-P2WPKH). By utilizing a local bitcoind node in regtest mode and automating the raw transaction workflow, this lab bridges theoretical blockchain concepts with command-line execution. Furthermore, the project seeks to analyze the underlying locking and unlocking scripts (the Challenge and Response) using the btcdeb debugger to verify execution flows. Ultimately, the goal is to perform a data-driven comparative analysis of these transaction structures. This comparison will mathematically demonstrate how the Segregated Witness upgrade reduces transaction fees through the "witness discount" and resolves third-party malleability to enable Layer 2 scaling solutions.

Part 1: Legacy Address Transactions (P2PKH)

Workflow Explanation

- **Network Setup:** Initialize a local Bitcoin node in regtest mode to create an isolated testing environment with standard fee configurations.
- **Address Creation & Funding:** Generate three standard Legacy addresses (A, B, and C). Fund Address A by mining blocks directly to it, providing the initial Unspent Transaction Output (UTXO) needed to start the chain.
- **First Transfer (A → B):** Construct a raw transaction to send funds from Address A to Address B. Sign the transaction using A's wallet and broadcast it to the local network.
- **Second Transfer (B → C):** Locate the newly created UTXO sitting at Address B. Use this specific UTXO as the input to build, sign, and broadcast a second raw transaction sending funds from Address B to Address C.
- **Script Analysis:** Decode both confirmed transactions to extract their raw metrics (Size, Vsize, Weight). Finally, isolate the locking (scriptPubKey) and unlocking (scriptSig) scripts and run them through the btcdeb debugger to verify how the legacy network processes public keys and signatures.

1. Transaction Workflow

Transaction 1 (Address A to Address B):

- **TXID:** c695f8000da38891413408f3ce116e1fbfed197490f8907ee5a71e5c2501c4b6

Transaction 2 (Address B to Address C):

- **TXID:** 2c5bfa73767f0d8245709b7db23bdea17ac97b321d89eb13273b9b865bdfa6eb

2. Decoded Scripts and Size Metrics

Transaction A to B

- **Size:** 225 bytes
- **Vsize:** 225 vbytes
- **Weight:** 900 WU
- **Locking Script (scriptPubKey.asm):** OP_DUP OP_HASH160 1bc9d39cdb4b8660d3586be4701c4225d9c94efd OP_EQUALVERIFY OP_CHECKSIG
- **Script Type:** pubkeyhash
- **Unlocking Script (scriptSig.asm):** 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a129f88b7e3[ALL] 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ bitcoin-cli -regtest decoderawtransaction "0200000001f25200cca478f871b51fc578114802cd30dbffffe4cee7a2079ae2cb9e5cd871010000006a47304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded139
c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a129f88b7e3012102dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45fdffffff0200c2eb0b000000001976a9141bc9d39cdb4b8660d3586be4701c4225d9c94efd88ac601ce011000000001976a914fb859ee62b42f
214e7fd1bf570ccf888ac00000000"
{
  "txid": "c695f8000da38891413408f3ce116e1fbfed197490f8907ee5a71e5c2501c4b6",
  "hash": "c695f8000da38891413408f3ce116e1fbfed197490f8907ee5a71e5c2501c4b6",
  "version": 2,
  "size": 225,
  "vsize": 225,
  "weight": 900,
  "locktime": 0,
  "vin": [
    {
      "txid": "71d85c9ecbe29a07a2e7cee4ffffdb30cd02481178c51fb571f878a4cc0052f2",
      "vout": 1,
      "scriptSig": {
        "asm": "304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a129f88b7e3[ALL] 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45",
        "hex": "47304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a129f88b7e3012102dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45"
      },
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 2.00000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "OP_DUP OP_HASH160 1bc9d39cdb4b8660d3586be4701c4225d9c94efd OP_EQUALVERIFY OP_CHECKSIG",
        "desc": "addr(mi3tKPhx68w6XD9D5CYK1QnkJ797wD8rBa)#7uewkkcq",
        "hex": "76a9141bc9d39cdb4b8660d3586be4701c4225d9c94efd88ac",
        "address": "mi3tKPhx68w6XD9D5CYK1QnkJ797wD8rBa",
        "type": "pubkeyhash"
      }
    },
    {
      "value": 2.99900000,
      "n": 1,
      "scriptPubKey": {
        "asm": "OP_DUP OP_HASH160 fb859ee62b42f3d514a16bf214e7fd1bf570ccf8 OP_EQUALVERIFY OP_CHECKSIG",
        "desc": "addr(n4SsyLEY1zKXWyoX792PJEmszcG2yawyWk)#pwr5gqzr",
        "hex": "76a914fb859ee62b42f3d514a16bf214e7fd1bf570ccf888ac",
        "address": "n4SsyLEY1zKXWyoX792PJEmszcG2yawyWk",
        "type": "pubkeyhash"
      }
    }
  ]
}
```

Transaction B to C

- **Size:** 225 bytes
- **Vsize:** 225 vbytes
- **Weight:** 900 WU
- **Locking Script (scriptPubKey.asm):** OP_DUP OP_HASH160 f992dfa35a97fc0cc081f50101975dc3274f7d97 OP_EQUALVERIFY OP_CHECKSIG
- **Script Type:** pubkeyhash
- **Unlocking Script (scriptSig.asm):** 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27f252b46f69a64e076b6f24fb347580eb9ceb1[ALL] 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ bitcoin-cli -regtest decoderawtransaction "0200000001b6c401255c1ea7e57e90f8907419edbf1f6e11cef30834419188a30d00f895c6000000006a47304022007780d6b756daad8586e9c0
02430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27f252b46f69a64e076b6f24fb347580eb9ceb101210388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9fdffffff0200e1f5050000000197
6a914f992dfa35a97fc0cc081f50101975dc3274f7d9788ac605af405000000001976a9141bc9d39cdb4b8660d3586be4701c4225d9c94efd88ac00000000"
{
  "txid": "2c5bfa73767f0d8245709b7db23bdea17ac97b321d89eb13273b9b865bdfa6eb",
  "hash": "2c5bfa73767f0d8245709b7db23bdea17ac97b321d89eb13273b9b865bdfa6eb",
  "version": 2,
  "size": 225,
  "vsize": 225,
  "weight": 900,
  "locktime": 0,
  "vin": [
    {
      "txid": "c695f8000da38891413408f3ce116e1fbfed197490f8907ee5a71e5c2501c4b6",
      "vout": 0,
      "scriptSig": {
        "asm": "3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27f252b46f69a64e076b6f24fb347580eb9ceb1[ALL] 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2
a91cd15f8eef44369ec9",
        "hex": "473044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27f252b46f69a64e076b6f24fb347580eb9ceb101210388a0f591b5c3576dbb9021623a2ef9635946d81049e2
a91cd15f8eef44369ec9"
      },
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 1.00000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "OP_DUP OP_HASH160 f992dfa35a97fc0cc081f50101975dc3274f7d97 OP_EQUALVERIFY OP_CHECKSIG",
        "desc": "addr(n4GaVmvxsqxGc1rCQ8GC4e3Uv3mtUvM3MQ)#xwcgk6rr",
        "hex": "76a914f992dfa35a97fc0cc081f50101975dc3274f7d9788ac",
        "address": "n4GaVmvxsqxGc1rCQ8GC4e3Uv3mtUvM3MQ",
        "type": "pubkeyhash"
      }
    },
    {
      "value": 0.99900000,
      "n": 1,
      "scriptPubKey": {
        "asm": "OP_DUP OP_HASH160 1bc9d39cdb4b8660d3586be4701c4225d9c94efd OP_EQUALVERIFY OP_CHECKSIG",
        "desc": "addr(mi3tKPhx68w6XD9D5CYKiQnkJ797wD8rBa)#7uewkkcq",
        "hex": "76a9141bc9d39cdb4b8660d3586be4701c4225d9c94efd88ac",
        "address": "mi3tKPhx68w6XD9D5CYKiQnkJ797wD8rBa",
        "type": "pubkeyhash"
      }
    }
  ]
}
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$
```

Script Validation Explanation: In a Legacy P2PKH transaction, the unlocking script (scriptSig / response) provides the user's cryptographic <Signature> and <Public Key>. The locking script (scriptPubKey / challenge) validates this by running OP_DUP to duplicate the public key, and OP_HASH160 to hash it. It then pushes the destination

address hash onto the stack and uses OP_EQUALVERIFY to ensure the hashes match. Finally, OP_CHECKSIG consumes the signature and public key to verify the mathematical proof of ownership.

Debugging transaction A->B

1) Initial State :

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ btcdeb '47304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec3
5685514eb95d4e7bb15fedca83a129f88b7e3012102dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c4576a914fb859ee62b42f3d514a16bf214e7fd1bf570ccf888ac'
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
7 op script loaded. type `help` for usage information
script                                                          |  stack
----------------------------------------------------------------+--------
304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d... |
02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45 |
OP_DUP                                                          |
OP_HASH160                                                      |
fb859ee62b42f3d514a16bf214e7fd1bf570ccf8                        |
OP_EQUALVERIFY                                                  |
OP_CHECKSIG                                                     |
#0000 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a129f88b7e301
btcdeb>
```

2) Mid-Execution :

```
btcdeb> step
                <> PUSH stack 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91ded13955d60220255c46f0a792e6fef1de050dec35685514eb95d4e7bb15fedca83a1
29f88b7e301
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45 | 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d...
OP_DUP                                                          |
OP_HASH160                                                      |
fb859ee62b42f3d514a16bf214e7fd1bf570ccf8                        |
OP_EQUALVERIFY                                                  |
OP_CHECKSIG                                                     |
#0001 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
btcdeb> step
                <> PUSH stack 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
OP_DUP                                                          | 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
OP_HASH160                                                      | 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d...
fb859ee62b42f3d514a16bf214e7fd1bf570ccf8                        |
OP_EQUALVERIFY                                                  |
OP_CHECKSIG                                                     |
#0002 OP_DUP
btcdeb>
```

```
btcdeb> step
                <> POP  stack
                <> PUSH stack fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
fb859ee62b42f3d514a16bf214e7fd1bf570ccf8                        |                                  fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
OP_EQUALVERIFY                                                  | 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
OP_CHECKSIG                                                     | 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d...
#0004 fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
btcdeb> step
                <> PUSH stack fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
OP_EQUALVERIFY                                                  |                                  fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
OP_CHECKSIG                                                     |                                  fb859ee62b42f3d514a16bf214e7fd1bf570ccf8
                                                                | 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
                                                                | 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d...

#0005 OP_EQUALVERIFY
```

3) Final State :

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                                          |                                                                stack
----------------------------------------------------------------+-----------------------------------------------------------------
OP_CHECKSIG                                                     | 02dfa619edabfe86cb8b02ceeba04832d8a066155cfd8b59c8de51052728d03c45
                                                                | 304402201ac16c3a607b06b075a15688d87c8f8b1b62bfe7e77a19cbc99d91d...

#0006 OP_CHECKSIG
btcdeb> step
EvalChecksig() sigversion=0
Eval Checksig Pre-Tapscript
error: Signature is found in scriptCode
btcdeb> ^C
```

Note on Final Execution: OP_EQUALVERIFY successfully matched the generated public key hash with the destination address hash. The final OP_CHECKSIG operation intentionally fails in this standalone btcdeb environment because the debugger lacks the full transaction context (inputs, outputs, amounts) required to independently calculate the transaction hash and verify the ECDSA signature mathematically.

Debugging transaction B->C

1) Initial State :

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ btcdeb '473044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27
f252b46f69a64e076b6f24fb347580eb9ceb101210388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec976a9141bc9d39cdb4b8660d3586be4701c4225d9c94efd88ac'
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
7 op script loaded. type `help` for usage information
script                                                          |  stack
----------------------------------------------------------------+--------
3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc... |
0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9 |
OP_DUP                                                          |
OP_HASH160                                                      |
1bc9d39cdb4b8660d3586be4701c4225d9c94efd                        |
OP_EQUALVERIFY                                                  |
OP_CHECKSIG                                                     |
#0000 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc71914c2f002207c52d9e6b836fdf4606cf978d27f252b46f69a64e076b6f24fb347580eb9ceb101
```

2) Mid-Execution :

```
btcdeb> step
                <> PUSH stack 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
script                                                          |                                                                stack
----------------------------------------------------------------+-----------------------------------------------------------------
OP_DUP                                                          | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
OP_HASH160                                                      | 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc...
1bc9d39cdb4b8660d3586be4701c4225d9c94efd                        |
OP_EQUALVERIFY                                                  |
OP_CHECKSIG                                                     |
#0002 OP_DUP
btcdeb> step
                <> PUSH stack 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
script                                                          |                                                                stack
----------------------------------------------------------------+-----------------------------------------------------------------
OP_HASH160                                                      | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
1bc9d39cdb4b8660d3586be4701c4225d9c94efd                        | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
OP_EQUALVERIFY                                                  | 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc...
OP_CHECKSIG                                                     |
#0003 OP_HASH160
```

```
btcdeb> step
                <> POP  stack
                <> PUSH stack 1bc9d39cdb4b8660d3586be4701c4225d9c94efd
script                                                          |                                                                stack
----------------------------------------------------------------+----------------------------------------------------------------
1bc9d39cdb4b8660d3586be4701c4225d9c94efd                        |                          1bc9d39cdb4b8660d3586be4701c4225d9c94efd
OP_EQUALVERIFY                                                  | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
OP_CHECKSIG                                                     | 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc...
#0004 1bc9d39cdb4b8660d3586be4701c4225d9c94efd
btcdeb> step
                <> PUSH stack 1bc9d39cdb4b8660d3586be4701c4225d9c94efd
script                                                          |                                                                stack
----------------------------------------------------------------+----------------------------------------------------------------
OP_EQUALVERIFY                                                  |                          1bc9d39cdb4b8660d3586be4701c4225d9c94efd
OP_CHECKSIG                                                     |                          1bc9d39cdb4b8660d3586be4701c4225d9c94efd
                                                                | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
                                                                | 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc...

#0005 OP_EQUALVERIFY
```

3) Final State :

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                                          |                                                                stack
----------------------------------------------------------------+----------------------------------------------------------------
OP_CHECKSIG                                                     | 0388a0f591b5c3576dbb9021623a2ef9635946d81049e2a91cd15f8eef44369ec9
                                                                | 3044022007780d6b756daad8586e9c002430408694671c41de8c4f15aecc8fc...

#0006 OP_CHECKSIG
btcdeb> step
EvalChecksig() sigversion=0
Eval Checksig Pre-Tapscript
error: Signature is found in scriptCode
btcdeb>
```

Part 2: SegWit P2SH-P2WPKH Transactions

Workflow Explanation

- **SegWit Address Creation:** Generate three new addresses (A', B', and C'), explicitly specifying the p2sh-segwit format so they utilize the Segregated Witness upgrade wrapped in backward-compatible P2SH formats.
- **Isolated Funding:** Fund Address A' by mining fresh blocks to it, ensuring these new SegWit transactions don't mix with the legacy UTXOs from Part 1.
- **First Transfer (A' → B'):** Just like in Part 1, construct, sign, and broadcast a raw transaction sending funds from Address A' to Address B'.
- **Second Transfer (B' → C'):** Identify the new P2SH-wrapped UTXO at Address B', and use it as the input to build, sign, and broadcast the final transaction to Address C'.
- **SegWit Analysis:** Decode these new transactions to observe the architectural changes. Specifically, document how the cryptographic signatures have been

moved out of the main script and into the txinwitness array, record the resulting drop in virtual size (vsize), and debug the outer P2SH wrapper to validate the transaction flow.

1. Transaction Workflow

Transaction 1 (Address A' to Address B'):

- **TXID:** ed1fe0c0c3549ae3fb13780032fc18787cd3c59e54cc0e5ac0a43f3fdf7cb889

Transaction 2 (Address B' to Address C'):

- **TXID:** b8179ad3fd20141f8e80d0c4304425c01e50a3ce123e035df52a878b20dbb623

2. Decoded Scripts and Size Metrics

Transaction A' to B'

- **Size:** 247 bytes
- **Vsize:** 166 vbytes
- **Weight:** 661 WU
- **Locking Script (scriptPubKey.asm):** OP_HASH160 947f6d2730bf924242c216be3b109d06a340bc5a OP_EQUAL
- **Script Type:** scripthash
- **Unlocking Script (scriptSig.asm):** 0014db01629ad94042e476184792f9384cb4d01540fd
- **Witness Data (txinwitness):**

  [0] (Signature): 3044022070081d32274c098a5d1ee23e042f377634e51b6a67e00c553ff9058fc1c4430c022060fc4b72cad3721636ed8f6e2575093e7c23e2951d06e561a1357f9c3f1a04f701

  [1] (PubKey): 02d2e6a313aa8474e88c7a41031d74cb16998bc9ceae45fcbd4614aa7abdac2eec

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ bitcoin-cli -regtest decoderawtransaction "020000000010108fe4ebb396addf176462c66e4ebb2bdb2e4ff6af344272d7e524d641fa7e2ba00000000171600 14db01629ad94042e476184792f9384cb4d01540fdfdffffff0200c2eb0b
0000000017a914947f6d2730bf924242c216be3b109d06a340bc5a87908f1b1e0100000017a914700f1e204b1aacf5bbdaebf55a575b594c9631bb870247304402207008 1d32274c098a5d1ee23e042f377634e51b6a67e00c553ff9058fc1c4430c022060fc4b72cad3721636ed8f6e2575093e7c23e2951d0
6e561a1357f9c3f1a04f701210 2d2e6a313aa8474e88c7a41031d74cb16998bc9ceae45fcbd4614aa7abdac2eec00000000"
{
  "txid": "ed1fe0c0c3549ae3fb13780032fc18707cd3c59e54cc0e5ac0a43f3fdf7cb889",
  "hash": "3963b0bd4c0308b6a558f8d8a656db512a672e9804b65fa3880934a8a388b83a",
  "version": 2,
  "size": 247,
  "vsize": 166,
  "weight": 661,
  "locktime": 0,
  "vin": [
    {
      "txid": "bae2a71f644d527e2d2744f36affe4b2bdb2ebe4662c4676f1dd6a39bb4efe08",
      "vout": 0,
      "scriptSig": {
        "asm": "0014db01629ad94042e476184792f9384cb4d01540fd",
        "hex": "160014db01629ad94042e476184792f9384cb4d01540fd"
      },
      "txinwitness": [
        "3044022070001d32274c098a5d1ee23e042f377634e51b6a67e00c553ff9058fc1c4430c022060fc4b72cad3721636ed8f6e2575093e7c23e2951d06e561a1357f9c3f1a04f701",
        "02d2e6a313aa8474e88c7a41031d74cb16998bc9ceae45fcbd4614aa7abdac2eec"
      ],
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 2.00000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "OP_HASH160 947f6d2730bf924242c216be3b109d06a340bc5a OP_EQUAL",
        "desc": "addr(2N6nQeR1yqeBVsEGQRTGNZQWsgR7i2a2Lxx)#kq7ut550",
        "hex": "a914947f6d2730bf924242c216be3b109d06a340bc5a87",
        "address": "2N6nQeR1yqeBVsEGQRTGNZQWsgR7i2a2Lxx",
        "type": "scripthash"
      }
    },
    {
      "value": 48.00090000,
      "n": 1,
      "scriptPubKey": {
        "asm": "OP_HASH160 700f1e204b1aacf5bbdaebf55a575b594c9631bb OP_EQUAL",
        "desc": "addr(2N3TjnATGfmE7HHpKFqu3ATF7xgupwa3mwf)#8zuel3gh",
        "hex": "a914700f1e204b1aacf5bbdaebf55a575b594c9631bb87",
        "address": "2N3TjnATGfmE7HHpKFqu3ATF7xgupwa3mwf",
        "type": "scripthash"
      }
    }
  ]
}
```

Transaction B' to C'

- **Size:** 247 bytes
- **Vsize:** 166 vbytes
- **Weight:** 661 WU
- **Locking Script (scriptPubKey.asm):** OP_HASH160 c6a04c30279056dc599d35416a377185e4b15462 OP_EQUAL
- **Script Type:** scripthash
- **Unlocking Script (scriptSig.asm):** 00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b
- **Witness Data (txinwitness):**

  [0] (Signature): 3044022067a7a075b88a4017f0472f6ab43eb703f600edce46f1fcad26acbb14808a5ec1022017bd004d8cef637ed262b3fcdd1c3ed238adb7cdd46a5489ca65eb251743e56201

  [1] (PubKey): 03dcc321f046cbdbee54e35fa93d44762314d74999813d127a8581143478986e74

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ bitcoin-cli -regtest decoderawtransaction "020000000010189b87cdf3f3fa4c05a0ecc549ec5d37c7818fc32007813fbe39a54c3c0e01fed00000000171600146f0901bf5ac91d6d6801c2dcf53cde72bc87174bfdffffff0200e1f505
0000000017a914c6a04c30279056dc599d35416a377185e4b1546287f0b9f5050000000017a914947f6d2730bf924242c216be3b109d06a340bc5a8702473044022067a7a075b88a4017f0472f6ab43eb703f600edce46f1fcad26acbb14808a5ec1022017bd004d8cef637ed262b3fcdd1c3ed238adb7cdd46
a5489ca65eb251743e56201210 3dcc321f046cbdbee54e35fa93d44762314d74999813d127a8581143478986e7400000000"
{
  "txid": "b8179ad3fd20141f8e80d0c4304425c01e50a3ce123e035df52a878b20dbb623",
  "hash": "57f66e1a01028d70b1727c64689a3613b002dd3d8171124aa2b6508c57929a9d",
  "version": 2,
  "size": 247,
  "vsize": 166,
  "weight": 661,
  "locktime": 0,
  "vin": [
    {
      "txid": "ed1fe0c0c3549ae3fb13780032fc18787cd3c59e54cc0e5ac0a43f3fdf7cb889",
      "vout": 0,
      "scriptSig": {
        "asm": "00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b",
        "hex": "1600146f0901bf5ac91d6d6801c2dcf53cde72bc87174b"
      },
      "txinwitness": [
        "304402206 7a7a075b88a4017f0472f6ab43eb703f600edce46f1fcad26acbb14808a5ec1022017bd004d8cef637ed262b3fcdd1c3ed238adb7cdd46a5489ca65eb251743e56201",
        "03dcc321f046cbdbee54e35fa93d44762314d74999813d127a8581143478986e74"
      ],
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 1.00000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "OP_HASH160 c6a04c30279056dc599d35416a377185e4b15462 OP_EQUAL",
        "desc": "addr(2NBMTowSvysAVVbQVQEQRNC6o5oHBm75VcP)#vt58vx4j",
        "hex": "a914c6a04c30279056dc599d35416a377185e4b1546287",
        "address": "2NBMTowSvysAVVbQVQEQRNC6o5oHBm75VcP",
        "type": "scripthash"
      }
    },
    {
      "value": 0.99990000,
      "n": 1,
      "scriptPubKey": {
        "asm": "OP_HASH160 947f6d2730bf924242c216be3b109d06a340bc5a OP_EQUAL",
        "desc": "addr(2N6nQeR1yqeBVsEGQRTGNZQWsgR71a2Lxx)#kq7ut550",
        "hex": "a914947f6d2730bf924242c216be3b109d06a340bc5a87",
        "address": "2N6nQeR1yqeBVsEGQRTGNZQWsgR71a2Lxx",
        "type": "scripthash"
      }
    }
  ]
```

Script Validation Explanation: In a P2SH-wrapped SegWit transaction, the heavy cryptographic signatures are removed from the main script. The unlocking script (scriptSig / response) only contains a lightweight <Witness Program>. The locking script (scriptPubKey / challenge) simply hashes this program using OP_HASH160 and verifies it matches the required hash using OP_EQUAL. Once this outer P2SH wrapper evaluates to true, upgraded SegWit nodes automatically evaluate the actual <Signature> and <Public Key> stored in the separated txinwitness structure.

Debugging transaction A' to B' :

1) Initial State :

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ btcdeb '160014db01629ad94042e476184792f9384cb4d01540fda914700f1e204b1aacf5bbdaebf
55a575b594c9631bb87'
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
4 op script loaded. type `help` for usage information
script                                        |  stack
----------------------------------------------+--------
0014db01629ad94042e476184792f9384cb4d01540fd |
OP_HASH160                                    |
700f1e204b1aacf5bbdaebf55a575b594c9631bb      |
OP_EQUAL                                      |
#0000 0014db01629ad94042e476184792f9384cb4d01540fd
btcdeb>
```

2) Validation :

```
btcdeb> step
                <> PUSH stack 0014db01629ad94042e476184792f9384cb4d01540fd
script                                        |                                        stack
----------------------------------------------+------------------------------------------
OP_HASH160                                    | 0014db01629ad94042e476184792f9384cb4d01540fd
700f1e204b1aacf5bbdaebf55a575b594c9631bb      |
OP_EQUAL                                      |
#0001 OP_HASH160
btcdeb> step
                <> POP  stack
                <> PUSH stack 700f1e204b1aacf5bbdaebf55a575b594c9631bb
script                                        |                                        stack
----------------------------------------------+------------------------------------------
700f1e204b1aacf5bbdaebf55a575b594c9631bb      |     700f1e204b1aacf5bbdaebf55a575b594c9631bb
OP_EQUAL                                      |
#0002 700f1e204b1aacf5bbdaebf55a575b594c9631bb
```

```
btcdeb> step
                <> PUSH stack 700f1e204b1aacf5bbdaebf55a575b594c9631bb
script                                        |                                        stack
----------------------------------------------+------------------------------------------
OP_EQUAL                                      |     700f1e204b1aacf5bbdaebf55a575b594c9631bb
                                              |     700f1e204b1aacf5bbdaebf55a575b594c9631bb
#0003 OP_EQUAL
```

3) Success :

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                        |                                        stack
----------------------------------------------+------------------------------------------
                                              |                                          01
```

Debugging transaction B' to C' :

1) Initial State :

```
sairaj@sairaj:~/Desktop/BlockChain/Assignment2$ btcdeb '1600146f0901bf5ac91d6d6801c2dcf53cde72bc87174ba914947f6d2730bf924242c216b
e3b109d06a340bc5a87'
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
4 op script loaded. type `help` for usage information
script                                        |  stack
----------------------------------------------+--------
00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b |
OP_HASH160                                    |
947f6d2730bf924242c216be3b109d06a340bc5a      |
OP_EQUAL                                      |
#0000 00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b
btcdeb>
```

2) Validation :

```
btcdeb> step
                <> PUSH stack 00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b
script                                      |                                       stack
--------------------------------------------+-----------------------------------------
OP_HASH160                                  | 00146f0901bf5ac91d6d6801c2dcf53cde72bc87174b
947f6d2730bf924242c216be3b109d06a340bc5a    |
OP_EQUAL                                    |
#0001 OP_HASH160
btcdeb> step
                <> POP  stack
                <> PUSH stack 947f6d2730bf924242c216be3b109d06a340bc5a
script                                      |                                       stack
--------------------------------------------+-----------------------------------------
947f6d2730bf924242c216be3b109d06a340bc5a    |     947f6d2730bf924242c216be3b109d06a340bc5a
OP_EQUAL                                    |
#0002 947f6d2730bf924242c216be3b109d06a340bc5a
```

```
btcdeb> step
                <> PUSH stack 947f6d2730bf924242c216be3b109d06a340bc5a
script                                      |                                       stack
--------------------------------------------+-----------------------------------------
OP_EQUAL                                    |     947f6d2730bf924242c216be3b109d06a340bc5a
                                            |     947f6d2730bf924242c216be3b109d06a340bc5a
#0003 OP_EQUAL
```

3) Success :

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                      |                                       stack
--------------------------------------------+-----------------------------------------
                                            |                                          01
```

Part 3: Comparative Analysis and Explanation

The following section provides a comprehensive comparative analysis between the Legacy Pay-to-Public-Key-Hash (P2PKH) transactions executed in Part 1 and the Pay-to-Script-Hash wrapped Segregated Witness (P2SH-P2WPKH) transactions executed in Part 2.

3.1 Comprehensive Size Metric Comparison

To understand the impact of the Segregated Witness (SegWit) upgrade, we must first look at the physical and virtual space each transaction occupies on the Bitcoin network. Below is the data extracted directly from our decoderawtransaction outputs.

Metric	Legacy P2PKH (Part 1 Avg)	SegWit P2SH-P2WPKH (Part 2 Avg)	Difference
Size (Raw Bytes)	225 bytes	247 bytes	+ 22 bytes (Larger)
Weight Units (WU)	900 WU	661 WU	- 239 WU (Lighter)
Virtual Size (vbytes)	225 vbytes	166 vbytes	**- 59 vbytes (Smaller)**

Analysis of the Metrics

At first glance, the data presents a fascinating anomaly: **the physical size of the SegWit transaction (247 bytes) is actually larger than the Legacy transaction (225 bytes), yet its Virtual Size (vsize) is significantly smaller (166 vbytes vs. 225 vbytes)**.

In the Legacy transaction, the size and vsize are exactly identical (225). This is because, prior to SegWit, the Bitcoin protocol calculated fees purely based on the raw byte size of the transaction string. However, SegWit introduced a new accounting mechanism based on "Weight Units" (WU) to decouple the physical size from the fee calculation, which fundamentally alters how the network measures the transaction.

*(Fee Savings Calculation: Savings = (1 - (SegWit vsize / Legacy vsize)) * 100% → (1 - (166 / 225)) * 100% = 26.22% fee reduction)*

3.2 Script Structure Comparison (Challenge & Response)

The primary reason for this discrepancy in sizing metrics lies in how the two transaction types construct their Challenge (Locking Script) and Response (Unlocking Script).

Legacy P2PKH Script Structure

In our Part 1 transactions, the entire cryptographic payload was stored directly inside the main script blocks.

- **The Challenge (scriptPubKey):** OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG
  - *Type:* pubkeyhash
  - *Mechanism:* This script demands that the spender provide a public key that hashes to the <pubKeyHash>, and a valid ECDSA signature corresponding to that public key.
- **The Response (scriptSig):** <Signature> <Public Key>
  - *Mechanism:* This script pushes the massive cryptographic signature (approx. 71-72 bytes) and the compressed public key (33 bytes) directly onto the execution stack.

Because these heavy cryptographic elements are placed directly inside the scriptSig, they are counted at full weight (4 WU per byte) by the legacy Bitcoin protocol, causing the transaction size and fee to bloat.

SegWit P2SH-P2WPKH Script Structure

In our Part 2 transactions, the architecture was fundamentally completely restructured to separate the signature from the script.

- **The Challenge (scriptPubKey):** OP_HASH160 <scriptHash> OP_EQUAL
  - *Type:* scripthash
  - *Mechanism:* Because we used a P2SH wrapper to ensure backward compatibility with older nodes, the locking script simply demands a hashed script that matches the destination <scriptHash>.
- **The Response (scriptSig):** <0014... witness program>
  - *Mechanism:* Noticeably absent from this unlocking script is the giant signature and public key! Instead, the scriptSig only contains a lightweight 22-byte "witness program" (0014db01629ad94042e476184792f9384cb4d01540fd in our A'->B' transaction). When hashed by the scriptPubKey, it satisfies the outer P2SH lock.
- **The Segregated Witness (txinwitness):** [<Signature>, <Public Key>]
  - *Mechanism:* The actual ECDSA signature and the public key have been segregated (moved outside) of the main script and placed into a completely new data structure called the txinwitness array.

3.3 Explanation: Why SegWit Transactions are "Smaller"

As noted in section 3.1, the SegWit transaction is physically larger (247 bytes vs 225 bytes) because it requires a few extra bytes to coordinate the P2SH wrapper and the new txinwitness flag. However, the Bitcoin network does not charge fees based on physical bytes anymore; it charges based on vsize.

SegWit transactions achieve a much smaller vsize due to the **Witness Discount**.

To calculate block limits and fees, Bitcoin applies a specific weight formula:

- **Non-Witness Data** (Transaction routing info, scriptPubKey, tiny P2SH scriptSig) is penalized at **4 Weight Units (WU) per byte**.
- **Witness Data** (The massive signatures and public keys inside the txinwitness array) is granted a 75% discount, counting as only **1 Weight Unit (WU) per byte**.

Because the heaviest part of any transaction is the cryptographic signature, moving it to the txinwitness array drastically reduces the total Weight of the transaction.

As seen in our raw data:

- Legacy Weight: **900 WU**
- SegWit Weight: **661 WU**

The Virtual Size is simply the total Weight divided by 4 $[vsize = ceil(weight/4)]$. Therefore, 661 / 4 = 165.25, which rounds up to our observed **166 vbytes**. This proves mathematically why SegWit transactions are treated as "smaller" by the network, despite containing slightly more physical bytes.

3.4 The Systemic Benefits of SegWit Transactions

The transition from Legacy P2PKH to SegWit architectures provides three critical benefits to the Bitcoin ecosystem:

1. Direct Economic Benefit to Users (Lower Fees)

Because miners calculate transaction fees strictly using the satoshis/vbyte metric, the mathematical reduction from 225 vbytes down to 166 vbytes translates directly into a ~26% discount on transaction fees for the user. By utilizing SegWit, users save money on every transfer.

2. Network Scalability and Throughput

Bitcoin blocks have a hard weight limit of 4,000,000 WU (which effectively replaced the old 1 Megabyte block size limit). By discounting the heaviest part of the transaction (the signatures), individual transactions consume far less of the block's maximum weight limit. This allows miners to physically pack significantly more transactions into a single

block, increasing the overall Transactions Per Second (TPS) throughput and capacity of the entire Bitcoin network without requiring a controversial hard fork.

3. Fixing Transaction Malleability (Enabling Layer 2)

While fee reduction is the most visible benefit, SegWit's most important technical achievement was solving **Third-Party Transaction Malleability**.

In Legacy P2PKH transactions (Part 1), the signature is located directly inside the scriptSig. Because the scriptSig is part of the data hashed to generate the overall Transaction ID (TXID), a malicious node could alter the cryptographic encoding of the signature slightly. This alteration wouldn't invalidate the signature, but it would change the resulting TXID before the block was mined.

This malleability made it impossible to build reliable chains of unconfirmed transactions, completely breaking Layer 2 scaling solutions like the **Lightning Network**. By explicitly segregating the signatures into the txinwitness array (Part 2), SegWit removes them from the TXID hash calculation. Consequently, signatures can no longer be maliciously altered to change the TXID, guaranteeing the stability required for the Lightning Network to function securely.

Conclusion

In conclusion, this assignment successfully demonstrates the structural and economic evolution of Bitcoin transactions from Legacy P2PKH to SegWit P2SH-P2WPKH formats. Through raw transaction decoding and script debugging, we verified that Legacy transactions store heavy cryptographic signatures directly within the main unlocking script (scriptSig), leading to higher physical weight and a critical vulnerability to transaction malleability.

Conversely, the implementation of SegWit effectively segregates these signatures into a completely separate txinwitness data structure. As proven by our metric analysis, this architectural shift grants a 75% weight discount on signature data, which resulted in a ~26% reduction in transaction fees (vsize) compared to our Legacy baseline. Beyond economic efficiency, moving the signatures outside of the main script permanently

resolves third-party malleability. By safeguarding the Transaction ID (TXID) from signature alterations, SegWit provides the essential foundational security required for unconfirmed transaction chains, allowing Layer 2 scaling solutions like the Lightning Network to function safely and efficiently.

References

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